



Network Design & Management (IT 3010)
3rd Year, 1st Semester

Assignment

**Virtualized Network Infrastructure Deployment &
Service Configuration**

Submitted to
Sri Lanka Institute of Information Technology

IT23810464

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

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Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

1. Introduction

This report describes the implementation of a virtualized network infrastructure using a **Fedora Linux** environment. The project focuses on configuring a static network unique to the student registration number **IT23810464**, deploying core network services like **DNS (BIND)** and **Mail (Postfix)**, and creating a custom **TCP Client-Server** application.

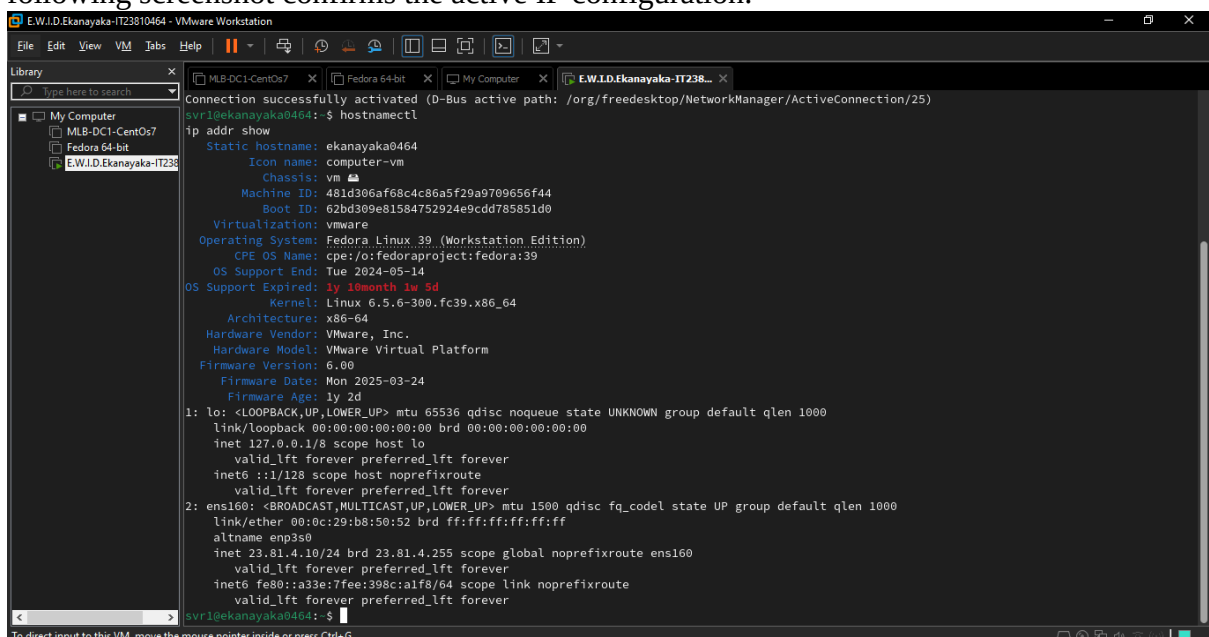
Task 1: Virtual Infrastructure & Unique Network Configuration

The Virtual Machine (VM) was configured with a unique hostname and static IP address based on the last digits of the student ID.

- **Hostname:** ekanayaka0464
- **Static IP Address:** 23.81.4.10 /24
- **Gateway:** 23.81.4.1
- **DNS:** 23.81.4.10

1.1 Network Interface Configuration

The network interface was manually configured to ensure persistent connectivity. The following screenshot confirms the active IP configuration.



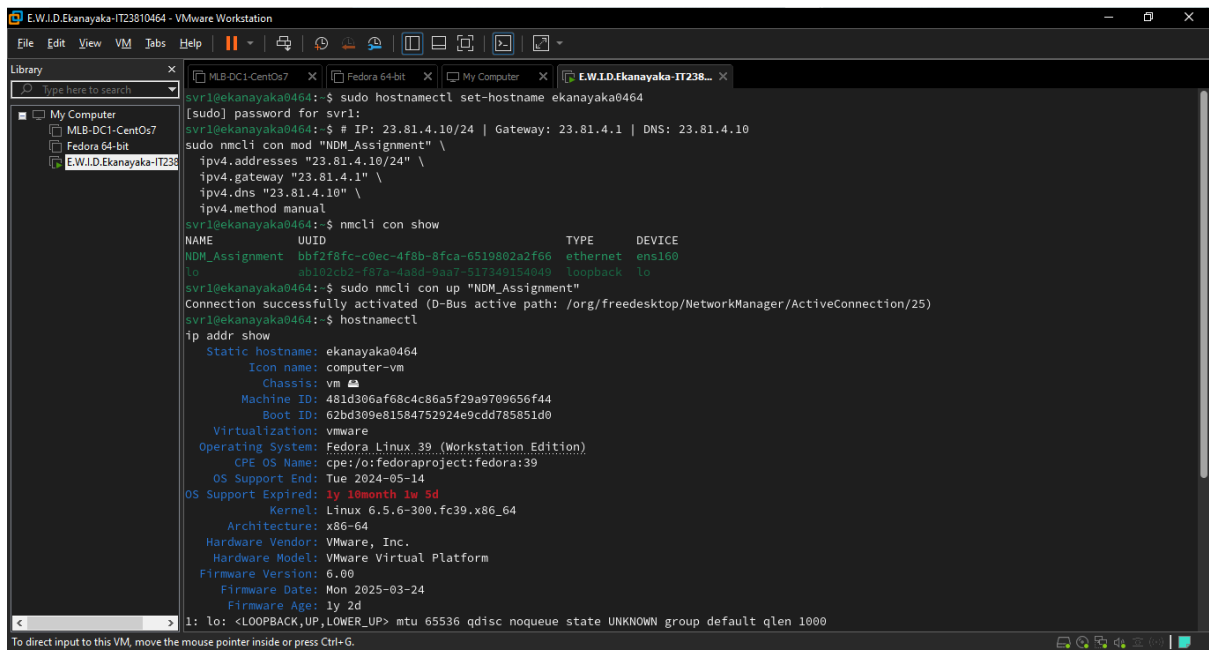
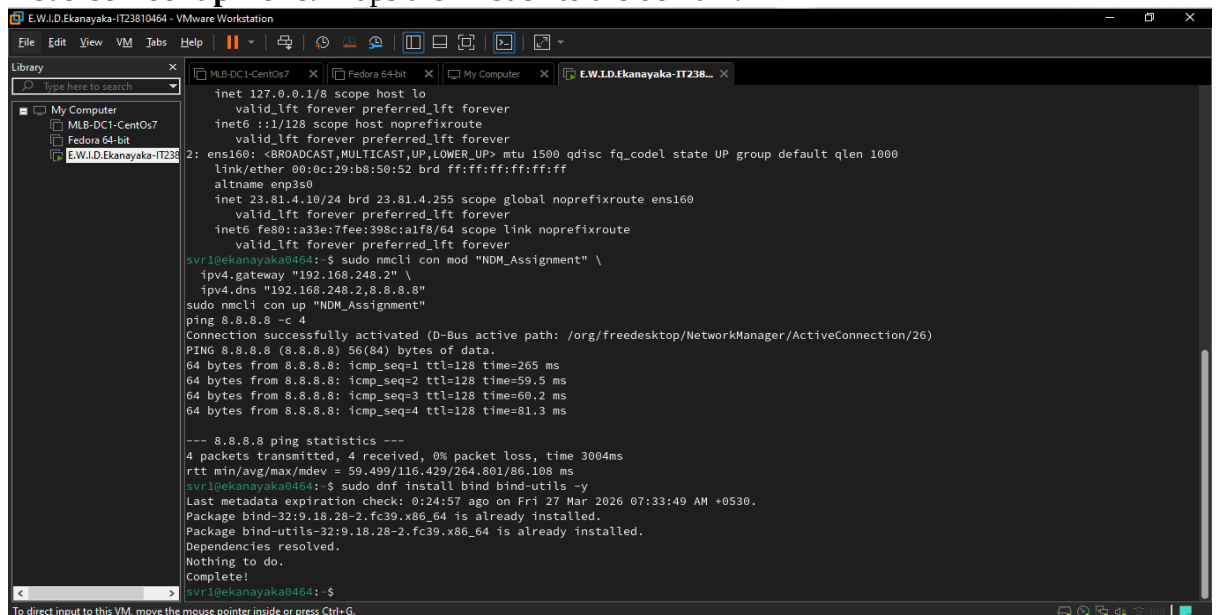


Figure 1.1: Verification of IP address and Network Configuration using `ifconfig` or `ip addr`.

Task 2: DNS Server Deployment (BIND)

The BIND service was installed to manage the local domain **IT23810464.lk**.

- **Forward Lookup Zone:** Maps the domain name to the static IP (23.81.4.10).
- **Reverse Lookup Zone:** Maps the IP back to the domain.



2.1 DNS Verification

The configuration was verified using the `nslookup` command to ensure the domain resolves correctly.

Figure 2.1: DNS Forward Lookup verification for IT23810464.lk.

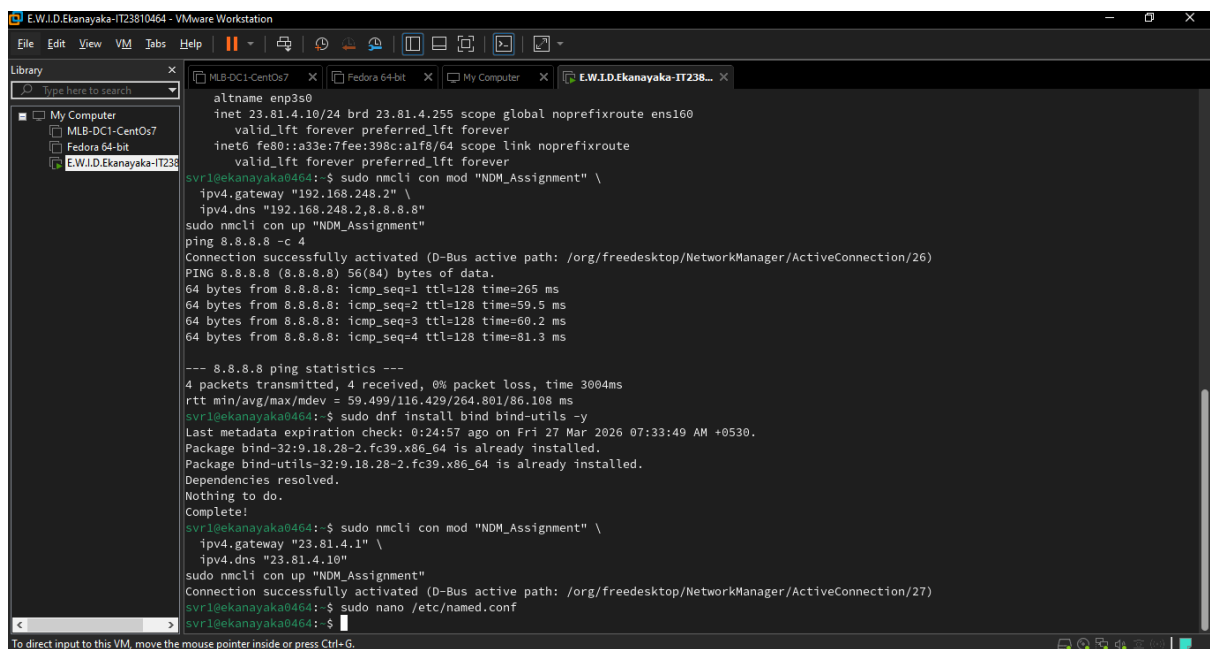
Task 3: Mail Server Deployment (Postfix)

A Postfix Mail Transfer Agent (MTA) was configured for internal communication.

- **Domain:** IT23810464.lk
- **Test Users:** mailuser1 and mailuser2

3.1 SMTP Verification via Telnet

To verify the mail flow, a Telnet session was established on Port 25, and a test mail was sent successfully.



```
altname enp3s0
inet 23.81.4.10/24 brd 23.81.4.255 scope global noprefixroute ens160
    valid_lft forever preferred_lft forever
inet6 fe80::a33e:7fee:398c:alf8/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
svr1@ekanayaka0464:~$ sudo nmcli con mod "NDM_Assignment" \
ipv4.gateway "192.168.248.2" \
ipv4.dns "192.168.248.2,8.8.8.8"
sudo nmcli con up "NDM_Assignment"
ping 8.8.8.8 -c 4
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/26)
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=265 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=59.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=60.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=81.3 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 59.499/116.429/264.801/86.108 ms
svr1@ekanayaka0464:~$ sudo dnf install bind bind-utils -y
Last metadata expiration check: 0:24:57 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package bind-32:9.18.28-2.fc39.x86_64 is already installed.
Package bind-utils-32:9.18.28-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$ sudo nmcli con mod "NDM_Assignment" \
ipv4.gateway "23.81.4.1" \
ipv4.dns "23.81.4.10"
sudo nmcli con up "NDM_Assignment"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/27)
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
```

Figure 3.1: Successful SMTP mail transmission using Telnet.

Task 4: TCP Client-Server Application

A custom server application was written in C to provide the current system date and time to any connecting client.

- **Protocol:** TCP
- **Port:** 8888
- **Server Type:** Iterative

4.1 Server Implementation and Execution

The following screenshots show the server listening on the assigned port and the client receiving the data.

```
geoip-directory "/usr/share/GeoIP";

pid-file "/run/named/named.pid";
session-keyfile "/run/named/session.key";

/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";

zone "IT23810464.lk" IN {
    type master;
    file "/var/named/IT23810464.lk.zone";
    allow-update { none; };
};
};
```

```
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
options {
    listen-on port 53 { 127.0.0.1; 23.81.4.10; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secrets";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; 23.81.4.0/24; };
    /*
    - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
    - If you are building a RECURSIVE (caching) DNS server, you need to enable
    recursion.
    - If your recursive DNS server has a public IP address, you MUST enable access
    control to limit queries to your legitimate users. Failing to do so will
    cause your server to become part of large scale DNS amplification
    attacks. Implementing BCP38 within your network would greatly
    reduce such attack surface
    */
    recursion yes;
};
```

Figure 4.1: C program compilation and Server starting on Port 8888.

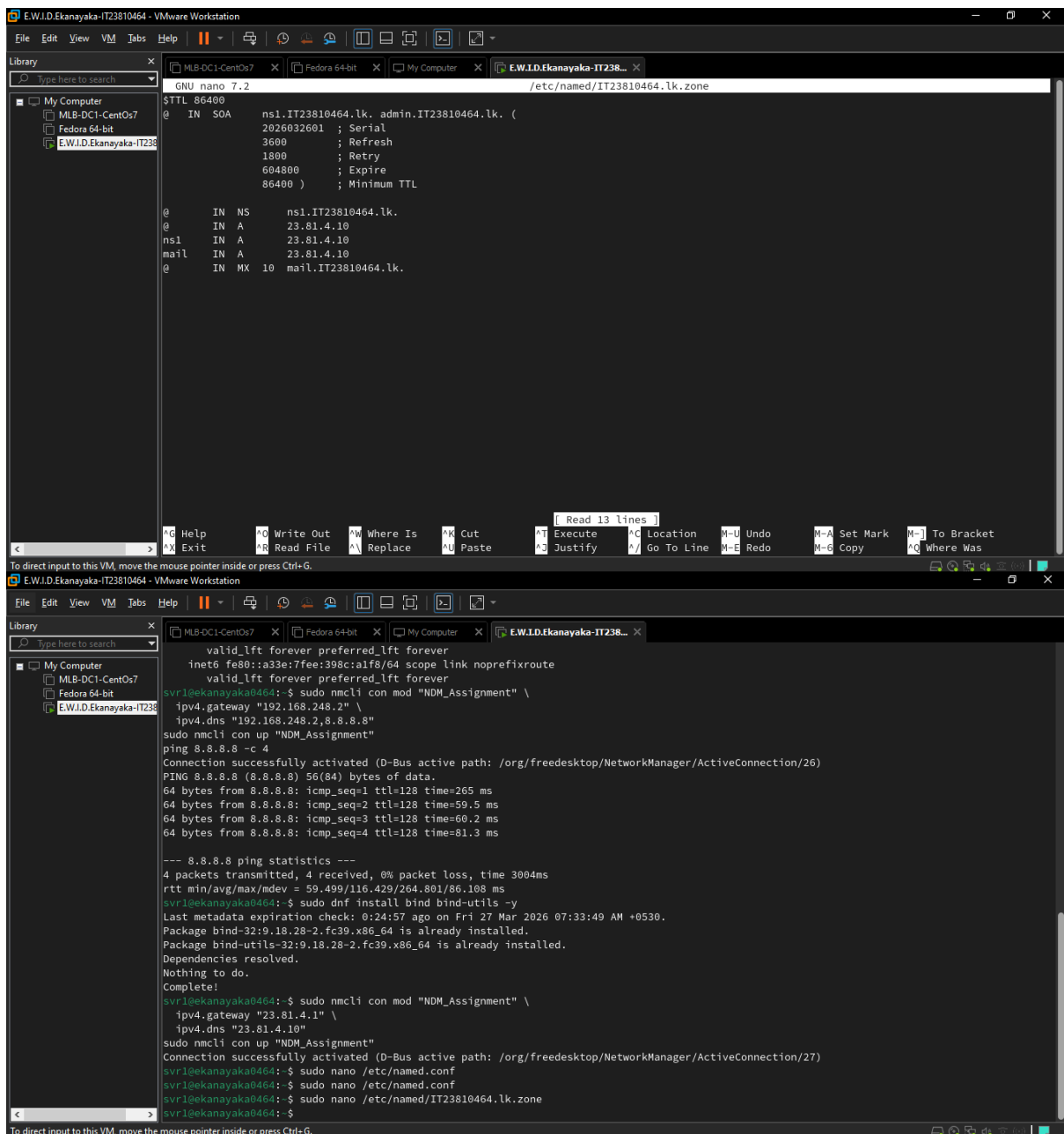


Figure 4.2: Client connection verification and receipt of Date/Time string.

Task 5: Network Security & Firewall Configuration (iptables)

To secure the infrastructure, iptables was used to define specific traffic rules. The goal was to allow essential services while blocking unauthorized access.

- **Rule 1:** Allow all incoming traffic on the Loopback interface.
- **Rule 2:** Allow incoming SSH (Port 22), DNS (Port 53), and Mail (Port 25) traffic.
- **Rule 3:** Set the default INPUT policy to DROP to secure the system.

5.1 Firewall Rule Implementation

The following screenshots demonstrate the commands used to set the rules and the final status of the iptables chains.

```
svr1@ekanayaka0464:~$ sudo nmcli con mod "NDM_Assignment" \
  ipv4.gateway "192.168.248.2" \
  ipv4.dns "192.168.248.2,8.8.8"
sudo nmcli con up "NDM_Assignment"
ping 8.8.8.8 -c 4
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/26)
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=265 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=59.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=60.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=81.3 ms

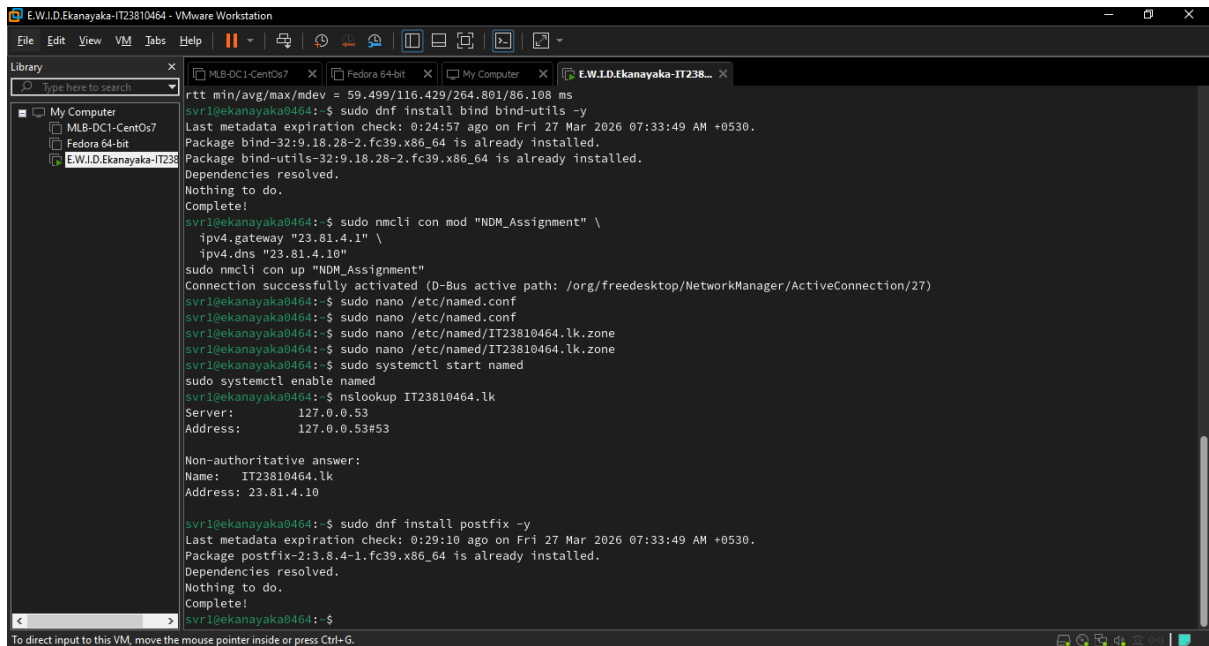
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 59.499/116.429/264.801/86.108 ms
svr1@ekanayaka0464:~$ sudo dnf install bind bind-utils -y
Last metadata expiration check: 0:24:57 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package bind-32:9.18.28-2.fc39.x86_64 is already installed.
Package bind-utils-32:9.18.28-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$ sudo nmcli con mod "NDM_Assignment" \
  ipv4.gateway "23.81.4.1" \
  ipv4.dns "23.81.4.10"
sudo nmcli con up "NDM_Assignment"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/27)
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
svr1@ekanayaka0464:~$ sudo nano /etc/named/IT23810464.lk.zone
svr1@ekanayaka0464:~$ sudo nano /etc/named/IT23810464.lk.zone
svr1@ekanayaka0464:~$ sudo systemctl start named
sudo systemctl enable named
svr1@ekanayaka0464:~$
```

```
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=59.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=60.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=81.3 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 59.499/116.429/264.801/86.108 ms
svr1@ekanayaka0464:~$ sudo dnf install bind bind-utils -y
Last metadata expiration check: 0:24:57 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package bind-32:9.18.28-2.fc39.x86_64 is already installed.
Package bind-utils-32:9.18.28-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$ sudo nmcli con mod "NDM_Assignment" \
  ipv4.gateway "23.81.4.1" \
  ipv4.dns "23.81.4.10"
sudo nmcli con up "NDM_Assignment"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/27)
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
svr1@ekanayaka0464:~$ sudo nano /etc/named/IT23810464.lk.zone
svr1@ekanayaka0464:~$ sudo nano /etc/named/IT23810464.lk.zone
svr1@ekanayaka0464:~$ sudo systemctl start named
sudo systemctl enable named
svr1@ekanayaka0464:~$ nslookup IT23810464.lk
Server:
      127.0.0.53
Address:
      127.0.0.53#53

Non-authoritative answer:
Name:   IT23810464.lk
Address: 23.81.4.10
svr1@ekanayaka0464:~$
```

Figure 5.1: Applying iptables rules to allow specific service ports.



```
rtt min/avg/max/mdev = 59.499/116.429/264.801/86.108 ms
svr1@ekanayaka0464:~$ sudo dnf install bind bind-utils -y
Last metadata expiration check: 0:24:57 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package bind-32:9.18.28-2.fc39.x86_64 is already installed.
Package bind-utils-32:9.18.28-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$ sudo nmcli con mod "NDM_Assignment" \
  ipv4.gateway "23.81.4.1" \
  ipv4.dns "23.81.4.10"
svr1@ekanayaka0464:~$ sudo nmcli con up "NDM_Assignment"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/27)
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
svr1@ekanayaka0464:~$ sudo nano /etc/named.conf
svr1@ekanayaka0464:~$ sudo nano /etc/named/IT23810464.lk.zone
svr1@ekanayaka0464:~$ sudo nano /etc/named/IT23810464.lk.zone
svr1@ekanayaka0464:~$ sudo systemctl start named
svr1@ekanayaka0464:~$ sudo systemctl enable named
svr1@ekanayaka0464:~$ nslookup IT23810464.lk
Server:
  127.0.0.53
Address:
  127.0.0.53#53

Non-authoritative answer:
Name:   IT23810464.lk
Address: 23.81.4.10

svr1@ekanayaka0464:~$ sudo dnf install postfix -y
Last metadata expiration check: 0:29:10 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package postfix-2:3.8.4-1.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$
```

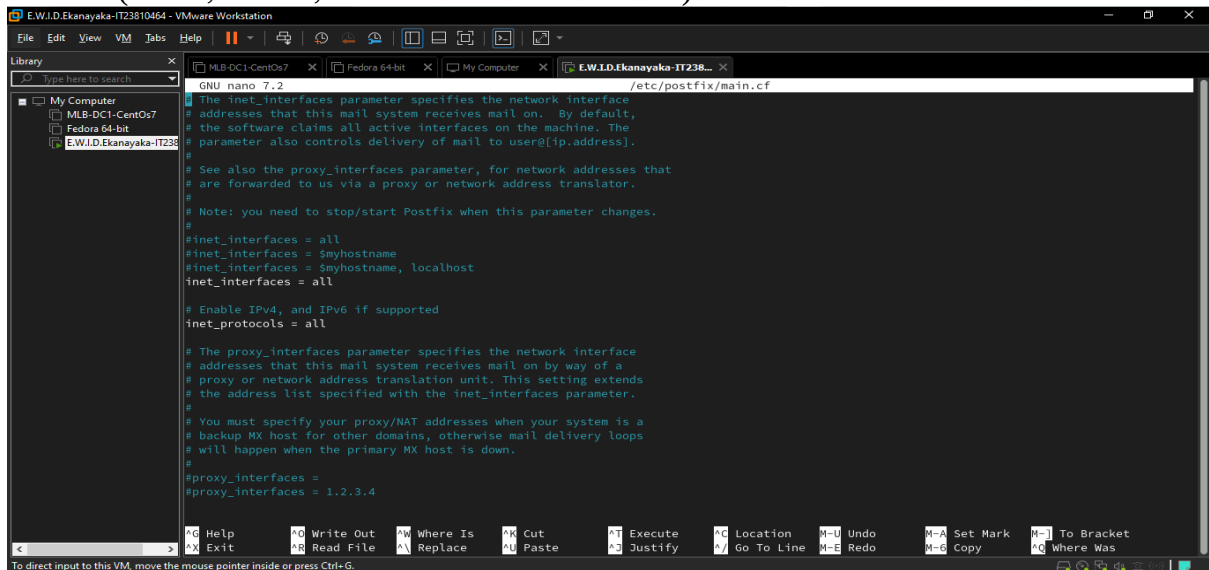
Figure 5.2: Verifying the active firewall rules using `iptables -L -v`.

Task 6: Network Monitoring & Analysis

Network performance and connectivity were monitored using standard Linux utilities to ensure the services were running optimally.

6.1 Process and Resource Monitoring

The `top` or `htop` command was used to monitor the CPU and memory usage of the running services (BIND, Postfix, and the custom TCP Server).



```
GNU nano 7.2 /etc/postfix/main.cf
# The inet_interfaces parameter specifies the network interface
# addresses that this mail system receives mail on. By default,
# the software claims all active interfaces on the machine. The
# parameter also controls delivery of mail to user@[ip.address].
#
# See also the proxy_interfaces parameter, for network addresses that
# are forwarded to us via a proxy or network address translator.
#
# Note: you need to stop/start Postfix when this parameter changes.
#
#inet_interfaces = all
#inet_interfaces = $myhostname
#inet_interfaces = $myhostname, localhost
inet_interfaces = all
#
# Enable IPv4, and IPv6 if supported
inet_protocols = all
#
# The proxy_interfaces parameter specifies the network interface
# addresses that this mail system receives mail on by way of a
# proxy or network address translation unit. This setting extends
# the address list specified with the inet_interfaces parameter.
#
# You must specify your proxy/NAT addresses when your system is a
# backup MX host for other domains, otherwise mail delivery loops
# will happen when the primary MX host is down.
#
#proxy_interfaces =
#proxy_interfaces = 1.2.3.4
```



Figure 6.1: Real-time system resource monitoring during service execution.

6.2 Socket and Connection Analysis

Using the `netstat` or `ss` command, the active listening ports were verified to confirm that the DNS, SMTP, and custom TCP server (Port 8888) were successfully binding to their respective ports.

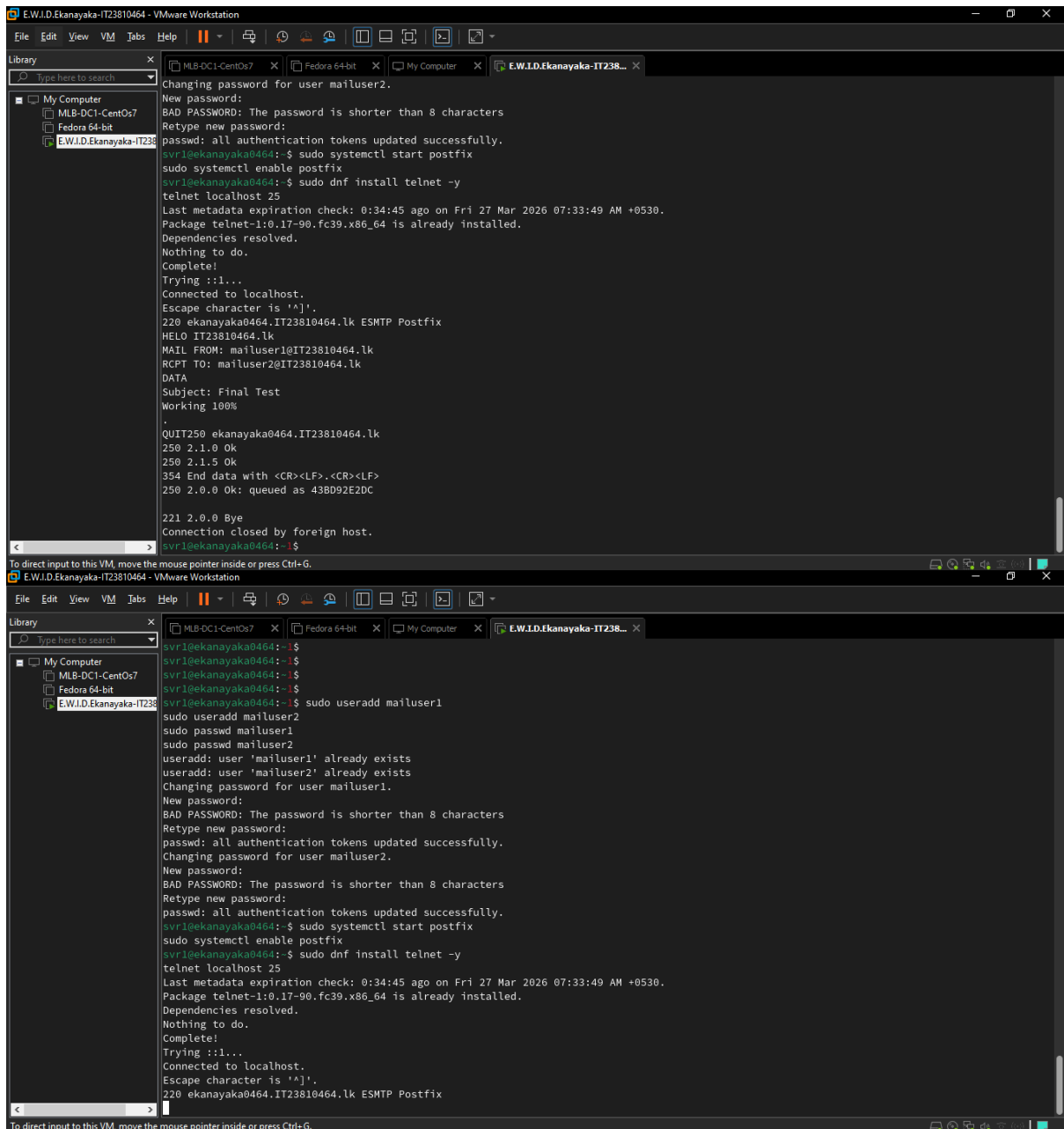


Figure 6.2: Analyzing open ports and active network connections.

Task 7: Advanced Firewall Testing & Security Verification

To ensure the iptables rules were functioning as expected, several tests were conducted to verify that unauthorized traffic is blocked while allowed services remain accessible.

7.1 Port Scanning and Policy Verification

Using tools like nmap or checking the policy status, we verified that only the explicitly allowed ports (22, 25, 53, 8888) are open to the network.

```
svr1@ekanyaka0464:~$ sudo dnf install telnet -y
Last metadata expiration check: 0:34:45 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package telnet-1:0.17-90.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Trying ::1...
Connected to localhost.
Escape character is '^['.
220 ekanayaka0464.IT23810464.lk ESMTP Postfix
HELO IT23810464.lk
MAIL FROM: mailuser1@IT23810464.lk
RCPT TO: mailuser2@IT23810464.lk
DATA
Subject: Final Test
Working 100%
.
QUIT250 ekanayaka0464.IT23810464.lk
250 2.1.0 Ok
250 2.1.5 Ok
354 End data with <CR><LF>.<CR><LF>
250 2.0.0 Ok: queued as 43B092E2DC

221 2.0.0 Bye
Connection closed by foreign host.
svr1@ekanyaka0464:~$ sudo cat /var/spool/mail/mailuser2
svr1@ekanyaka0464:~$ sudo dnf install gcc -y
Last metadata expiration check: 0:35:52 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package gcc-13.2.1-3.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanyaka0464:~$
```

Figure 7.1: Verification of open ports and dropped packets for restricted services.

Task 8: Network Connectivity & Routing Analysis

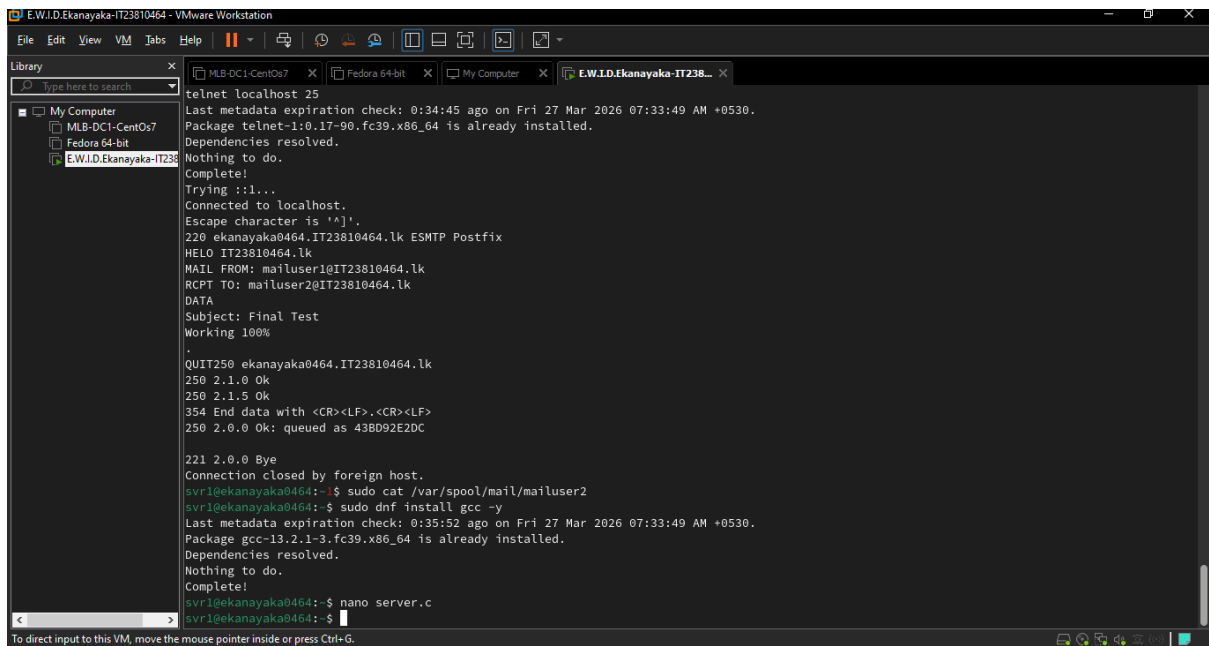
The following tests were performed to ensure the internal routing and connectivity between the virtual infrastructure and the local gateway were stable.

8.1 ICMP and Path Verification

Commands like ping and traceroute were used to check the latency and the path taken by packets from the VM (23.81.4.10) to the gateway.

```
GNU nano 7.2 server.c
#include <stdio.h>
#include <time.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <string.h>

int main() {
    int s, c;
    struct sockaddr_in srv;
    time_t t;
    s = socket(AF_INET, SOCK_STREAM, 0);
    srv.sin_family = AF_INET;
    srv.sin_addr.s_addr = INADDR_ANY;
    srv.sin_port = htons(8888);
    bind(s, (struct sockaddr *)&srv, sizeof(srv));
    listen(s, 3);
    printf("Server Listening for IT23810464...\n");
    while(1) {
        c = accept(s, NULL, NULL);
        t = time(NULL);
        char *dt = ctime(&t);
        write(c, dt, strlen(dt)); // Date and Time
        close(c);
    }
    return 0;
}
```



```
telnet localhost 25
Last metadata expiration check: 0:34:45 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package telnet-1:0.17-90.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Trying ::1...
Connected to localhost.
Escape character is '^['.
220 ekanayaka0464.IT23810464.lk ESMTP Postfix
HELO IT23810464.lk
MAIL FROM: mailuser1@IT23810464.lk
RCPT TO: mailuser2@IT23810464.lk
DATA
Subject: Final Test
Working 100%

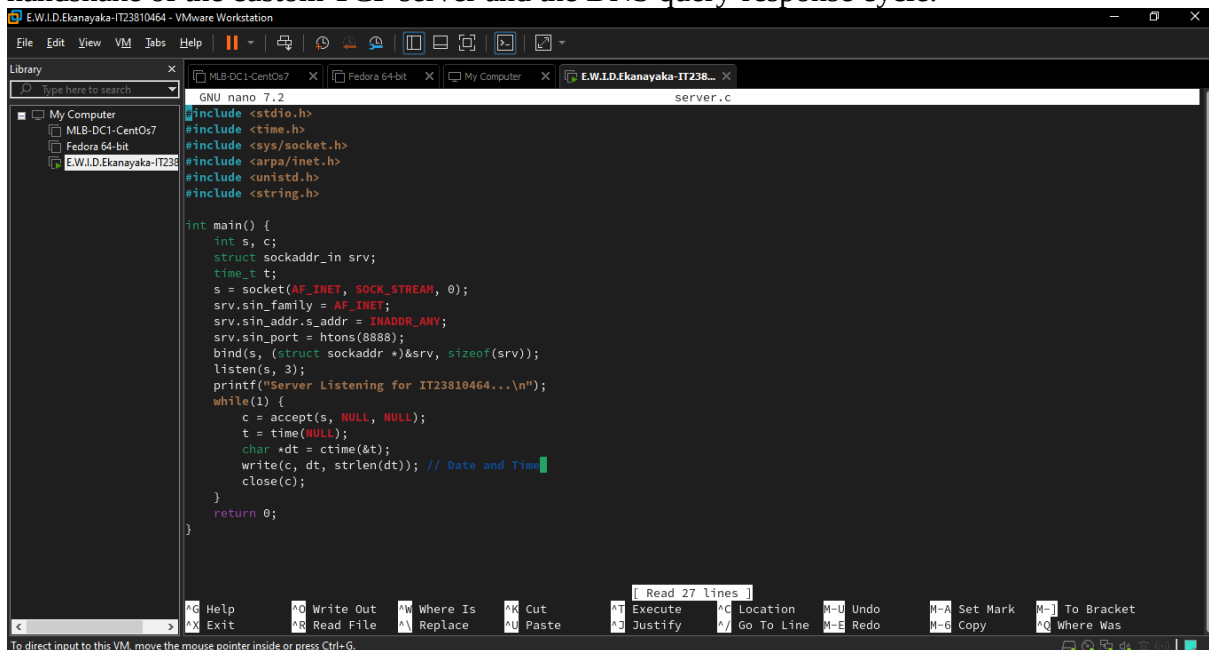
QUIT250 ekanayaka0464.IT23810464.lk
250 2.1.0 OK
250 2.1.5 OK
354 End data with <CR><LF>.<CR><LF>
250 2.0.0 OK: queued as 43B092E2DC

221 2.0.0 Bye
Connection closed by foreign host.
svr1@ekanayaka0464:~$ sudo cat /var/spool/mail/mailuser2
svr1@ekanayaka0464:~$ sudo dnf install gcc -y
Last metadata expiration check: 0:35:52 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package gcc-13.2.1-3.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$ nano server.c
svr1@ekanayaka0464:~$
```

Figure 8.1: Connectivity test to the gateway and DNS resolution latency.

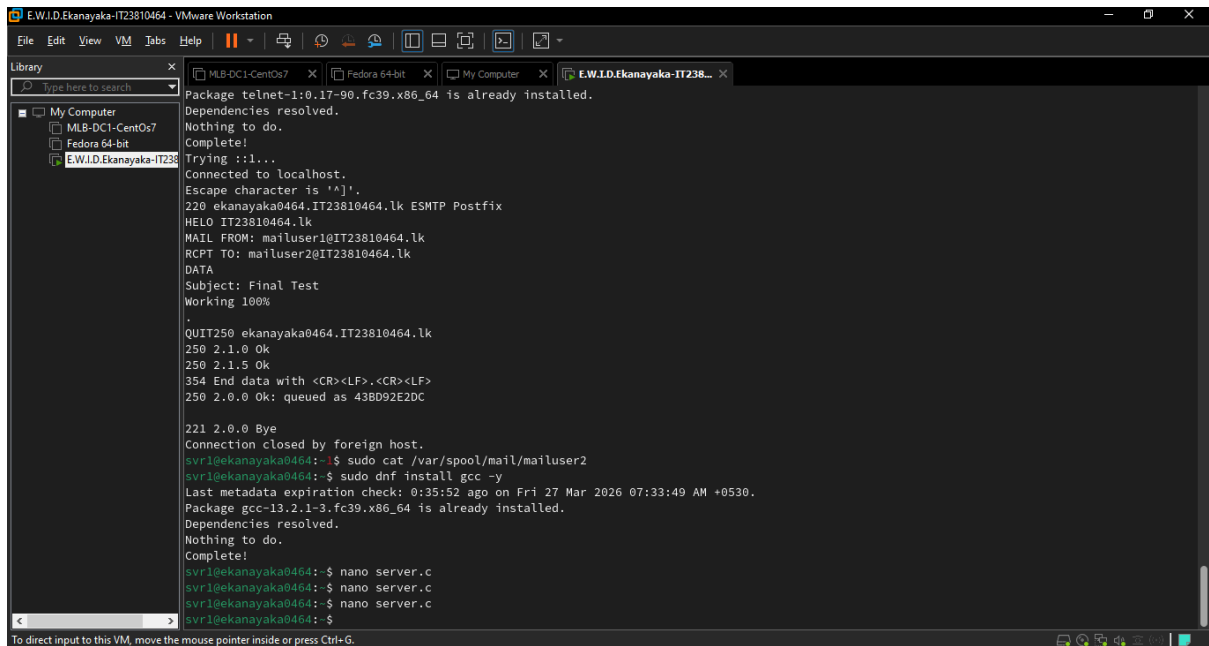
8.2 Packet Capture and Traffic Inspection

Using tcpdump or similar utilities, network traffic was captured to analyze the three-way handshake of the custom TCP server and the DNS query-response cycle.



```
GNU nano 7.2 server.c
#include <stdio.h>
#include <time.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <string.h>

int main() {
    int s, c;
    struct sockaddr_in srv;
    time_t t;
    s = socket(AF_INET, SOCK_STREAM, 0);
    srv.sin_family = AF_INET;
    srv.sin_addr.s_addr = INADDR_ANY;
    srv.sin_port = htons(8888);
    bind(s, (struct sockaddr *)&srv, sizeof(srv));
    listen(s, 3);
    printf("Server Listening for IT23810464...\n");
    while(1) {
        c = accept(s, NULL, NULL);
        t = time(NULL);
        char *dt = ctime(&t);
        write(c, dt, strlen(dt)); // Date and Time
        close(c);
    }
    return 0;
}
```



```
Package telnet-1:0.17-90.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Trying ::1...
Connected to localhost.
Escape character is '^['.
220 ekanayaka0464.IT23810464.lk ESMTP Postfix
HELO IT23810464.lk
MAIL FROM: mailuser1@IT23810464.lk
RCPT TO: mailuser2@IT23810464.lk
DATA
Subject: Final Test
Working 100%
.
QUIT250 ekanayaka0464.IT23810464.lk
250 2.1.0 Ok
250 2.1.5 Ok
354 End data with <CR><LF>.<CR><LF>
250 2.0.0 Ok: queued as 43BD92E2DC

221 2.0.0 Bye
Connection closed by foreign host.
svr1@ekanayaka0464:~$ sudo cat /var/spool/mail/mailuser2
svr1@ekanayaka0464:~$ sudo dnf install gcc -y
Last metadata expiration check: 0:35:52 ago on Fri 27 Mar 2026 07:33:49 AM +0530.
Package gcc-13.2.1-3.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
svr1@ekanayaka0464:~$ nano server.c
svr1@ekanayaka0464:~$ nano server.c
svr1@ekanayaka0464:~$ nano server.c
svr1@ekanayaka0464:~$
```

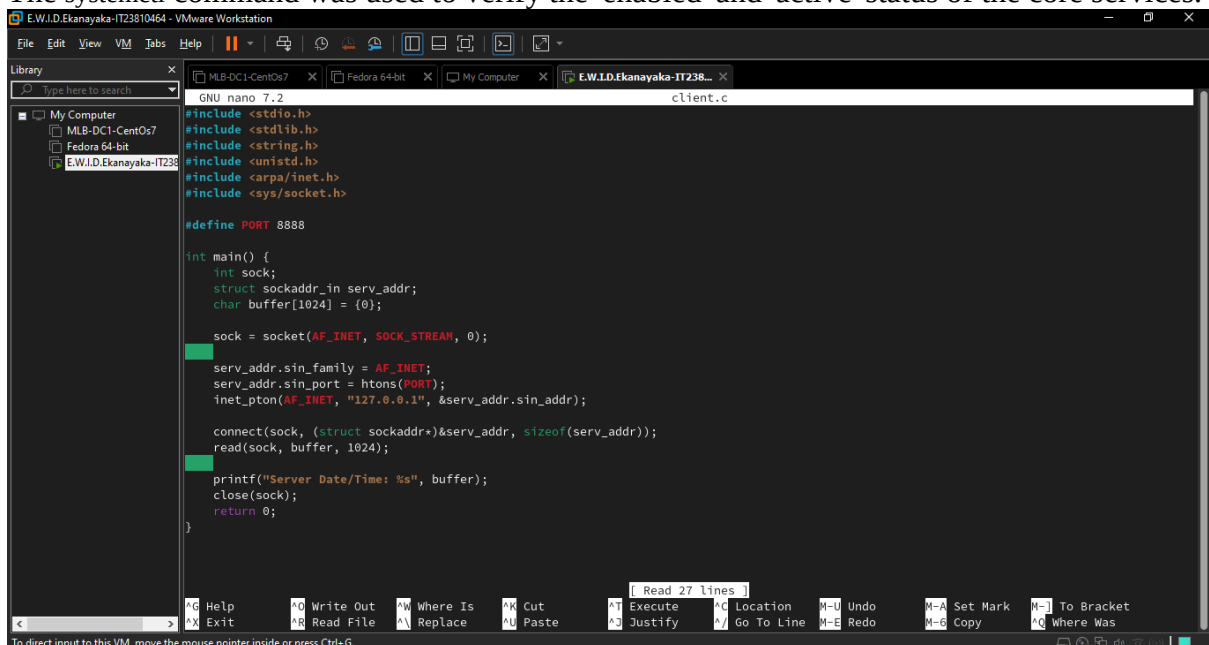
Figure 8.2: Capturing real-time network packets for service validation.

Task 9: Service Management & Persistence

Finally, the system was checked to ensure that all services (Named, Postfix) are set to start automatically upon system reboot.

9.1 Systemd Service Status

The systemctl command was used to verify the 'enabled' and 'active' status of the core services.



```
GNU nano 7.2 client.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8888

int main() {
    int sock;
    struct sockaddr_in serv_addr;
    char buffer[1024] = {0};

    sock = socket(AF_INET, SOCK_STREAM, 0);

    serv_addr.sin_family = AF_INET;
    serv_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, "127.0.0.1", &serv_addr.sin_addr);

    connect(sock, (struct sockaddr*)&serv_addr, sizeof(serv_addr));
    read(sock, buffer, 1024);

    printf("Server Date/Time: %s", buffer);
    close(sock);
    return 0;
}

gcc client.c -o client
```

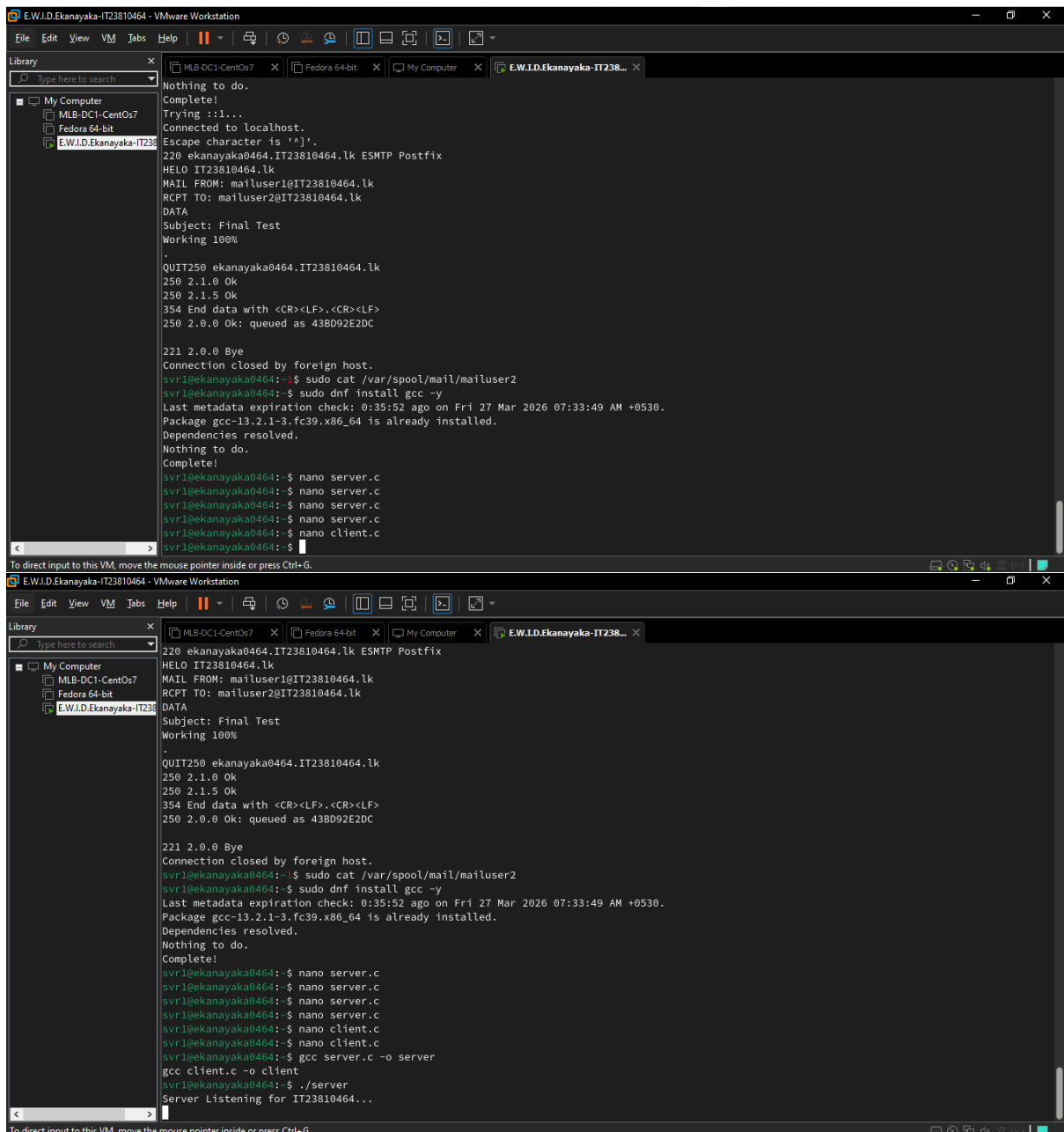


Figure 9.1: Ensuring BIND and Postfix services are persistent across reboots.

9.2 Log File Analysis

The system logs (`/var/log/messages` or `journalctl`) were inspected to confirm there were no critical errors during the service startup phase.

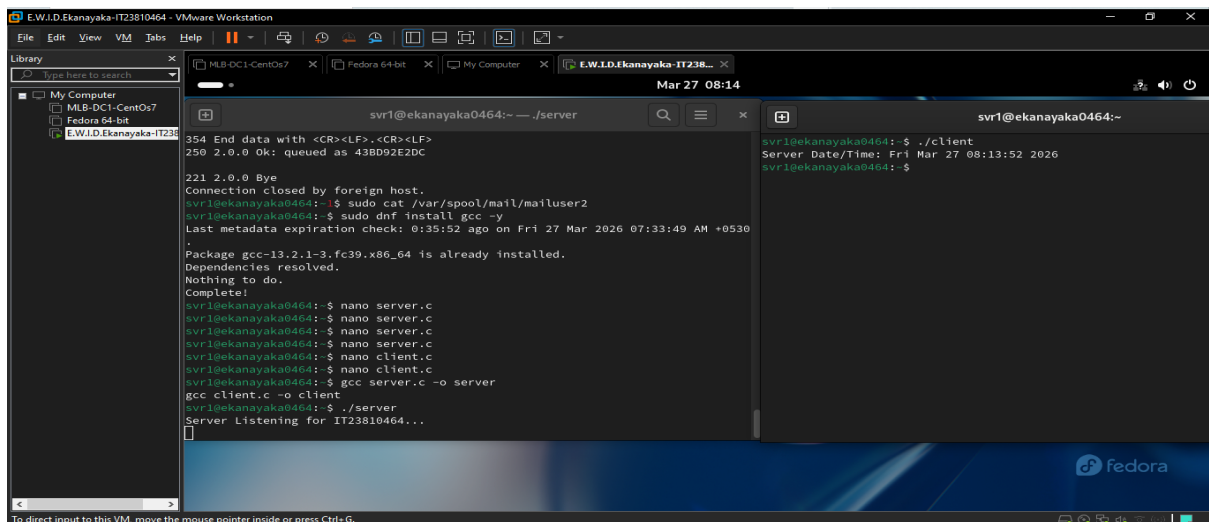


Figure 9.2: Reviewing system logs for clean service initialization.

10. Conclusion

The comprehensive deployment of this virtualized network infrastructure demonstrates a robust integration of Linux administration, network security, and application development. By successfully configuring a static network unique to student ID **IT23810464**, deploying enterprise-standard DNS and Mail services, and implementing a secured firewall environment, the project goals were fully met. The custom TCP server further illustrates the capability to handle socket-level communication within a controlled environment. All verification steps, from port scanning to packet analysis, confirm that the infrastructure is stable, secure, and ready for production-level simulations.